Q1. JEE Main 2026 (28 January Shift 2)

The correct order of acidic strength of the major products formed in the given reactions, is :

A. $PhNH_2 \xrightarrow[\substack{(2)\ CuCN \\ (3)\ H_3O^+/\Delta}]{(1)\ NaNO_2\ +\ HCl\ (<5°C)} [A]$

B. $CH_3CH_2CHO \xrightarrow[\Delta]{[Ag(NH_3)_2]^+, OH^-} [B]$

C. $CH_4 + O_2 \xrightarrow[\text{(ii) } Na_2Cr_2O_7/H^+]{\text{(i)}} [C]$

D. $PhCH_2MgBr + CO_2 \xrightarrow[H_3O^+]{\text{Dry ether}} [D]$

Choose the correct answer from the options given below :

(1) $A > D > B > C$

(2) $C > A > D > B$

(3) $C > B > A > D$

(4) $A > D > C > B$

ANSWERS AND SOLUTIONS

1. (2)

1. **(2)** Product [A]: From $PhNH_2$, steps (1) $NaNO_2$ + HCl forms diazonium salt, (2) CuCN gives benzonitrile, (3) H_3O^+/Δ hydrolyzes to benzoic acid (PhCOOH), pKa ≈ 4.2.
Product [B]: From CH_3CH_2CHO, Tollens' oxidation with $[Ag(NH_3)_2]^+$ produces acetic acid (CH_3COOH), pKa ≈ 4.8.
Product [C]: From $CH_4 + O_2$ with heat and $Na_2Cr_2O_7/H^+$ produces formic acid (HCOOH), pKa ≈ 3.8 (strongest acid).
Product [D]: From $PhCH_2MgBr + CO_2$, the Grignard reaction produces phenylacetic acid ($PhCH_2COOH$), pKa ≈ 4.3.
Acidic strength order (lower pKa = stronger): C (3.8) > A (4.2) > D (4.3) > B (4.8).